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GNNDRP: Graph neural network with multi-task learning for drug response prediction

Congzhi Song, Xuan Liu, Zhankun Xiong, Luotao Liu and Wen Zhang

Abstract—Using computational methods to personalize drug response prediction holds great promise to improve cancer therapy. Most existing methods use either biochemical information or response-related networks to predict drug response, nevertheless, the information they considered is not comprehensive. In this study, we present a novel end-to-end deep learning-based method **G**raph **N**eural **N**etwork with multi-task learning for **D**rug **R**esponse **P**rediction (GNNDRP). It leverages biochemical features as well as the hidden features from the heterogeneous network which incorporates the known drug-cell line responses, drug similarities, and cell line similarities, to complete the drug response prediction task. Moreover, GNNDRP designs a self-supervised task to enhance the representation capacity from the response network and further improve the model prediction performance. Extensive experiments show that GNNDRP outperforms existing state-of-the-art prediction methods under various experimental settings. The ablation analysis reveals that the biochemical characteristics, response-related network, and our self-supervised strategy can boost the predictive power. Additionally, case studies further validate the effectiveness of GNNDRP in identifying novel drug-cell line responses.

Index Terms—Drug response prediction, Deep learning, Multi-task learning, Self-supervised learning.

1 INTRODUCTION

DUE to significant advances in high-throughput molecular profiling technologies, precision medicine has gained much attention in recent years [1]. The object of precision medicine is to design treatments specifically adapted to the characteristics of a particular molecular subgroup of tumors or even individual patients by profiling tumors at the molecular level. Precision medicine needs extensive experimental patient drug screening projects, which are performed to discover significant patterns of response. However, the traditional experimental verification of the response between a large number of patients and drugs is time-consuming, expensive, and impractical [2]. To tackle this barrier, cancer cell lines extracted from clinical tumor samples have been used as a surrogate of patients.

In the past decades, several large-scale high-throughput screening efforts have been made to catalog genomic information of a panel of in vitro cancer cell lines and their drug sensitivity profiles against anticancer compounds. Cancer Cell Line Encyclopedia (CCLE) [3] and Genomics of Drug Sensitivity in Cancer (GDSC) [4] are the most famous databases in this field. The response values of thousands of cancer cell lines against hundreds of anticancer compounds and cancer cell lines' molecular profiles are recorded in these databases. These valuable information resources provide researchers with a golden opportunity to understand the mechanism of cancer treatment, and have been extensively utilized to establish computational methods for drug response prediction.

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A variety of computational methods have been proposed for predicting drug response. Network-based methods treat drug response prediction as a link prediction task in the known drug-cell line response network or similarity network [5]–[9]. Classification-based methods use different approaches to process the biochemical characteristics of drugs and cell lines as the feature vectors, then train a classifier, such as random forest [10], [11], logistic regression [12], and SVM [13], [14], to predict drug response. Moreover, matrix factorization [15], [16] methods are also used for developing novel drug response predictive methods, these methods are capable of learning interactions among features from different views by the decomposed factors. Recently, deep learning has been widely used in many computational biology tasks and achieves excellent performance [17]–[20], especially in the drug response prediction problem [21]–[23]. For instance, CDRscan [21] uses the ensemble CNN model to process the molecular fingerprint of drugs and genomic mutation of cell lines for drug response prediction. DeepDSC [22] employs a stacked deep autoencoder to encode gene expression data, the encoded features are merged with precomputed molecular fingerprints and then feed into a fully connected predictive network. It has been demonstrated that graph neural networks (GNNs) can process graph-structured data more outstandingly as a popular technology in the deep learning area. Instead of traditionally using convolution operation to process the fingerprints of drugs, several GNN-based methods [24]–[26] employ graph neural networks to learn the drug structure representation from drug molecular graphs and have achieved significant progress in drug response prediction.

These pioneering methods obtain excellent results, however, there still exists room to further improve the predictive performance. Considering the cancer drug response can be converted into a drug-cell line interaction network and GNNs have demonstrated convincing performance on

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many biochemical network analyses [27], [28], it motivates us to develop a GNN-based method to boost prediction performance. GNNs mainly utilize unlabeled nodes by simply aggregating their features that cannot thoroughly take advantage of the abundant information [29]. To fully exploit the unlabeled nodes, self-supervised learning (SSL) can be naturally harnessed for providing additional supervision. According to [30], self-supervised mechanisms can be roughly divided into three schemes, 1) pre-training & fine-tuning, 2) self-training and 3) multi-task learning. Since multi-task learning has a good performance and general framework among the three, we choose this type of self-supervised learning.

In this work, we develop a novel end-to-end deep learning-based method, namely **G**raph **N**eural **N**etwork with multi-task learning for **D**rug **R**esponse **P**rediction (GNNDRP), which integrates both biochemical characteristics and response-related network information to predict drug response. We first use graph isomorphism networks to extract drug structure features and utilize 1-D CNNs to learn the genomic mutation feature of cell lines. Second, we construct a heterogeneous network based on the known drug-cell line response network, drug-drug similarity network, and cell line-cell line similarity network, then use the graph attention network to learn the response-related network features for drugs and cell lines. Third, randomly covering features is elaborately adopted for the response-related network learning, which is a specific self-supervised strategy and further improves the predictive performance. Finally, two types of features of drugs and cell lines are concatenated to get the fusion features, then feed into a multi-layer perceptron (MLP) and obtain the prediction score. Massive experiments show that our method outperforms other state-of-the-art methods. The ablation analysis reveals that integrating two types of features and utilizing the SSL strategy can enhance prediction performance.

The main contributions of this work are as follows:

- GNNDRP completes the drug response prediction task by utilizing more comprehensive information from drug molecular structures, cell line genomic expressions, and drug-cell line response network.
- GNNDRP adopts a specific self-supervised task to enhance the representation learning capacity from the drug-cell line response network and further improve the model's predictive performance.
- GNNDRP is a novel end-to-end deep learning-based method that incorporates both drug response prediction task and self-supervised task into a multi-task framework, achieving more outstanding performance than other state-of-the-art methods under various experimental settings.

2 MATERIALS AND METHODS

2.1 Data preparation

In this study, we collect the drug response data, cell line genomic mutation data from the GDSC database [4], and drug SMILES [31] information from the PubChem database [32].

For the drugs, GDSC database provides their names and the compound id (CID) as their identity number. We remove the repeating drugs in the GDSC database, then discard the drugs that CID did not match the PubChem dataset. At last, 231 drugs and their corresponding SMILES structural information are collected. For the cell lines, genomic characteristics of 990 cell lines are obtained from the GDSC database, including mutation status and copy number alterations.

GDSC database also provides IC_{50} values across hundreds of drugs and cell lines. We collect the IC_{50} values for measuring drug's effectiveness against cancer cell line, then perform the logarithmic operation on the IC_{50} value and transform the relationship between the drug and the cell line into a binary state (sensitive or resistant) by setting thresholds provided by [4]. Finally, we obtain 185,769 drug-cell line response pairs, of which 2,1399 are sensitive and 16,4370 are resistant. Considering all the 231 x 990 = 228,690 drug-cell line response pairs, approximately 18.8% (42,921) of the IC_{50} values are missing/unknown.

2.2 Graph neural network with multi-task learning for drug response prediction

As shown in Fig.1, GNNDRP is composed of four modules: drug structure feature learning module, cell line mutation feature learning module, self-supervised graph learning on heterogeneous network module, and drug response prediction module. For a given drug-cell line pair, the biochemical characteristics of the drug and cell line are obtained through the drug structure feature learning module and cell line mutation feature learning module, respectively. A graph neural network with self-supervised learning is proposed to get the response network features. Finally, two types of features of drugs and cell lines are concatenated and fed into the drug response prediction module.

2.2.1 Drug structure feature learning

The SMILES of drugs can be converted into a molecular graph with RDKit [33], where the nodes of the graph represent atoms, and the edges represent chemical bonds between atoms. Formally, for a given drug u, the molecular graph is described as $\mathcal{G}_u = (\boldsymbol{X}_u, \boldsymbol{A}_u)$, where $\boldsymbol{X}_u \in \mathbb{R}^{N_u \times C}$ and $\boldsymbol{A}_u \in \mathbb{R}^{N_u \times N_u}$ represent the node feature matrix and adjacency matrix respectively. N_u is the number of atoms of drug u and C is the dimension of feature vector. Following the description in [25], the attributes of a atom are represented as a fixed dimensional vector ($C = 78$), including the atom symbol, the degree of the atom, the total number of the Hydrogen, the implicit value, and whether it is an aromatic compound.

We adopt Graph Isomorphism Network (GIN) [34], which has proven its maximum discriminative power among GNNs [34], to process the molecular graph of drugs. The GIN layer using multi-layer perceptron to update the feature vector is defined as:

$$\boldsymbol{H}_u^{(l+1)} = MLP^{(l+1)}((\boldsymbol{A}_u + (1+\epsilon^{(l+1)}) \cdot \boldsymbol{I}) \cdot \boldsymbol{H}_u^{(l)}) \quad (1)$$

where $\boldsymbol{H}_u^{(l)}$ is the node feature matrix of drug u at the l-th layer and ϵ is a learnable parameter or a fixed scalar. We initialize $\boldsymbol{H}_u^{(0)} = \boldsymbol{X}_u$.

ReLU activation function and batch normalization is used in our model after each GIN layer. We apply a global max pooling over all nodes to produce a summary representation vector of the entire molecular graph. At last, a fully connected layer is utilized to transform the embedding dimension. As a result, each drug is represented by a vector with 128 dimensions.

2.2.2 Cell line mutation feature learning

The genomic features of cell lines can be represented as a binary sequence. Similar to [35], 1-D convolutional neural networks are used to learn latent features of a given cell line v as:

$$f_{CNN}: \boldsymbol{x}_v \in \mathbb{R}^{1\times d_c} \mapsto \boldsymbol{y}_v \in \mathbb{R}^{1\times d_c^{'}} \tag{2}$$

where d_c and $d_c^{'}$ are the dimension of the genomic feature and latent representation of the cell line v, respectively. ReLU activation function and max-pooling techniques are adopted after each convolution operation. Finally, a fully connected layer is also adopted and each cell line is represented by a 128-dimensional vector.

2.2.3 Self-supervised graph learning on heterogeneous network

Heterogeneous network. The heterogeneous network we constructed is composed of three parts: drug-drug similarity network, cell line-cell line similarity network, and drug-cell line response network. Specifically, the heterogeneous graph $\boldsymbol{G} \in \mathbb{R}^{(n+m)\times(n+m)}$ is defined by the adjacency matrix:

$$\boldsymbol{G} = \begin{vmatrix} \boldsymbol{S}_d & \boldsymbol{A} \\ \boldsymbol{A}^T & \boldsymbol{S}_c \end{vmatrix} \tag{3}$$

where $\boldsymbol{A} \in \mathbb{R}^{n\times m}$ is the adjacency matrix which contains sensitive linkages between drugs and cell lines, $\boldsymbol{S}_d \in \mathbb{R}^{n\times n}$ and $\boldsymbol{S}_c \in \mathbb{R}^{m\times m}$ denote drug-drug similarity matrix and cell line-cell line similarity matrix, respectively. The drug-drug similarity matrix is built by using similarity information of drug-pairs. Each drug is converted into a binary sequence of the same dimension by computing their Morgan fingerprint, and then the Cosine similarity calculation method is used to obtain the drug similarity matrix. The Cosine similarity measure is also adopted to calculate the cell line similarity matrix by using genomic profiles of cell lines.

Then, the feature matrix $\boldsymbol{H} = \{\boldsymbol{h}_1, \boldsymbol{h}_2, ..., \boldsymbol{h}_{n+m}\} \in \mathbb{R}^{(n+m)\times(n+m)}$ is initialized as:

$$\boldsymbol{H} = \begin{vmatrix} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^T & \boldsymbol{0} \end{vmatrix} \tag{4}$$

Graph encoder. Graph attention network (GAT) leveraging self-attention mechanism for learning low-dimensional representations of nodes from graph-structured data [36]. In this study, we use GAT layers with multi-head attention mechanism as aggregation operators for node features update which is defined as:

$$\boldsymbol{z}_i = \|_{k=1}^{K} \sigma\Big(\sum_{j\in\mathcal{N}_{(i)}} \alpha_{ij}^k \boldsymbol{W}^k \boldsymbol{h}_j\Big) \tag{5}$$

where $\alpha_{ij}^k \in \mathbb{R}$ is the attention coefficient by the k-th attention mechanism, $\boldsymbol{W}^k$ is a learnable transformation matrix, $\mathcal{N}_{(i)}$ is the first-order neighbors of node i (including i), $||$ represents concatenate operation, K indicates the multi-head numbers, and σ is a non-linear activation function which is set to ELU following [36]. The final node embedding matrix $\boldsymbol{Z} = \{\boldsymbol{z}_1, \boldsymbol{z}_2, ..., \boldsymbol{z}_{n+m}\} \in \mathbb{R}^{(n+m)\times d}$ is obtained

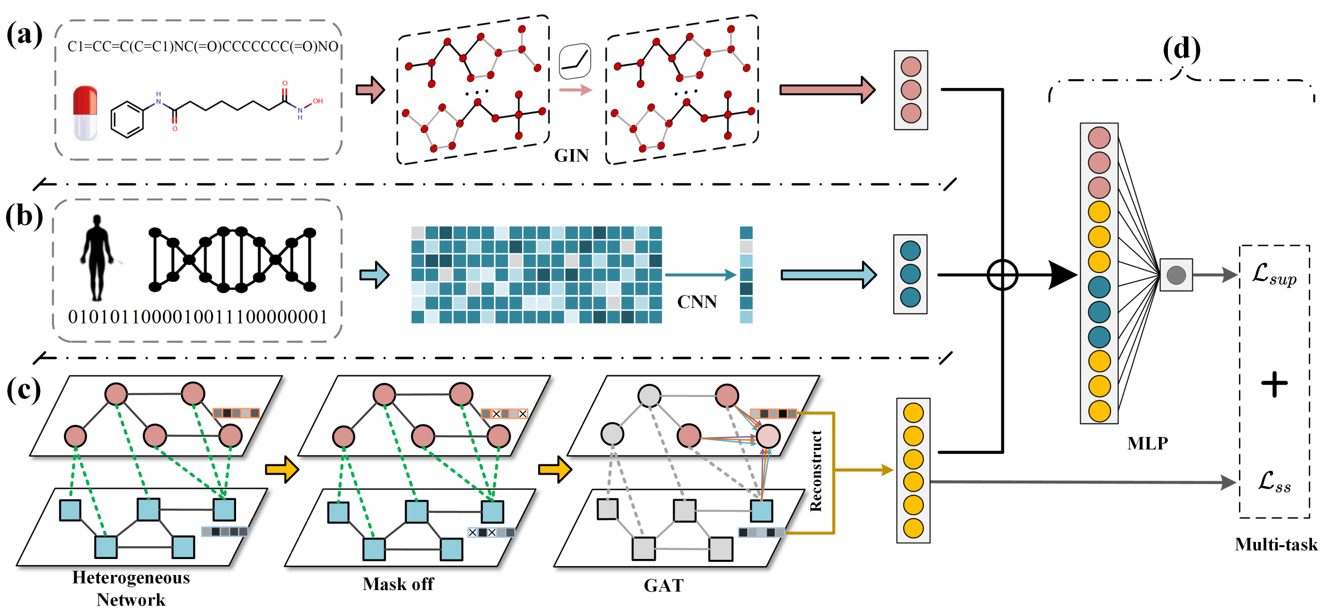

Fig. 1. Overview of GNNDRP framework. (**a**) Drug structure features are extracted by the GIN encoder from the molecular graph. (**b**) Cell line mutation features are generated by the CNN encoder from the gene expression profiles. (**c**) Response-related network features are learned by a GAT encoder with a self-supervised task from a heterogeneous network. In the heterogeneous network, the red and blue nodes denote drugs and cell lines, respectively, and the solid and dashed edges denote similarity and drug-cell line responses, respectively. The mask off operation means randomly covering the initial feature of nodes. (**d**) These two types of features (biochemical features and response-related network features) are combined by concatenate operation and then feed into an MLP decoder to get the final prediction score.

via multi-head GAT layers, where d is the output dimension of the last GAT layer. The coefficient α_{ij} is expressed as:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\boldsymbol{a}^T\left[\boldsymbol{W}\boldsymbol{h}_i \| \boldsymbol{W}\boldsymbol{h}_j\right]\right)\right)}{\sum_{k\in\mathcal{N}_{(i)}}\exp\left(\text{LeakyReLU}\left(\boldsymbol{a}^T\left[\boldsymbol{W}\boldsymbol{h}_i \| \boldsymbol{W}\boldsymbol{h}_k\right]\right)\right)} \tag{6}$$

where $\boldsymbol{a}^T$ is a weight vector.

Self-supervised learning task. Self-supervised learning (SSL) automatically generates supervision signals by studying inherent information. We design an SSL task that adopts the randomly covering features strategy on the node feature matrix to make the graph encoder learn the node representation. Specifically, the initial feature vectors of nodes are randomly covered with the mask-ratio p_{mask}. It then aims at recovering masked node features by feeding to graph encoder unmasked node features. The self-supervised learning process is achieved by the link prediction task. The objective of this task is to reconstruct an adjacency matrix $\hat{\boldsymbol{A}}$ close to true one $\boldsymbol{A}$:

$$\hat{\boldsymbol{A}} = sigmoid(\hat{\boldsymbol{Z}}_D \hat{\boldsymbol{Z}}_C^T) \tag{7}$$

where $\hat{\boldsymbol{Z}}_D \in \mathbb{R}^{n\times d}$ and $\hat{\boldsymbol{Z}}_C \in \mathbb{R}^{m\times d}$ is the final feature matrix using SSL strategy of drug and cell line, respectively. For self-supervised task, we adopt the binary cross entropy as:

$$\mathcal{L}_{ss} = -\frac{1}{|o^+| + |o^-|}\left(\sum_{(i,j)\in o^+}\log(\hat{A}_{ij}) + \sum_{(i,j)\in o^-}\log\left(1-\hat{A}_{ij}\right)\right) \tag{8}$$

where o^+ and o^- are the positive set and negative set in masked adjacency matrix, respectively.

2.2.4 Drug response prediction

For a given drug-cell line pair (u, v) through the preceding module, we can obtain the drug structure feature, the cell line mutation feature, and the response-related network features for both drug and cell line. Then the high-level feature of the drug and cell line are concatenated and fed into an MLP to yield the final prediction score $\hat{r}_{uv}$. The objective function of classification task is:

$$\mathcal{L}_{\text{sup}} = -\frac{1}{|p^+| + |p^-|}\left(\sum_{(i,j)\in p^+}\log\left(\hat{r}_{ij}\right) + \sum_{(i,j)\in p^-}\log\left(1-\hat{r}_{ij}\right)\right) \tag{9}$$

where p^+ and p^- denote the set of sensitive instances and the set of resistant instances, respectively.

Considering both supervised task and self-supervised task, the final loss function in GNNDRP is:

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \alpha\mathcal{L}_{ss} \tag{10}$$

where α is the weight factor that is used to control the impact of $\mathcal{L}_{ss}$. During training, we learn the model parameter by minimizing the total loss $\mathcal{L}$ using Adam optimizer [37].

3 Experiments

3.1 Experimental setting

To evaluate the predictive ability of the proposed method, we adopt the 5-fold cross-validation (5-CV) scheme. The dataset is split into five equal-sized subsets. This process is repeated five times, and every subset is used as the testing set in turn while the remaining four subsets are used as the training set. In terms of evaluation metrics, the area under the receiver-operating characteristic curve (AUC), and the area under the accuracy-recall rate curve (AUPR) are chosen in our experiment to measure the performance of the models. We demonstrate results under various experimental settings in the following:

- **Rediscovering known drug-cell line responses.** Based on the known 185,769 drug-cell line interactions across 231 drugs and 990 cell lines, we randomly select 80% of instances as the training set and the remaining 20% of instances as the testing set for model evaluation (keep the consistency of the ratio between sensitive pairs and resistant pairs).
- **Blind test for both drugs and cell lines.** Considering the condition when given a new drug or new cell line that is not included in the training data. Therefore, we design the blind test to further evaluate the predictive power of our method. We randomly split drugs/cell lines into five subsets of equal size. In each fold, we choose one subset of drugs/cell lines using them to simulate newly emerged drugs/cell lines for testing, and collect the other four subsets of drugs/cell lines using them to train the prediction model, and then make predictions for the new ones.

For the drug structure feature learning module, three GIN layers are stacked to build GIN architecture. For the cell line mutation feature learning module, the 1-D CNNs with three layers is chosen as the function f_{CNN}. For the self-supervised graph learning on the heterogeneous network module, we use a two-layer GAT model. The first layer consists of $K = 4$ attention heads, and the second layer uses a single attention head where the output dimension d is set to 128. For the drug response prediction module, the five-layer MLP with 256, 128, 64, 32, and 1 hidden units is adopted to conduct the final prediction, while ELU and Sigmoid are the activation function of the former four layers and the last layer, respectively. The initial learning rate of the optimizer is set to 0.001 and the total training epochs of GNNDRP are set to 1000. Furthermore, there are two important parameters that influence the performance of GNNDRP: the mask ratio p_{mask} and the loss function weight α. We consider these parameters from the ranges $p_{mask} \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6\}$ and $\alpha \in \{0.1, 0.2, 0.3, ..., 1.0\}$. By adjusting the parameters empirically, we set the $p_{mask} = 0.2$ and $\alpha = 0.5$ for GNNDRP in the following experiments.

3.2 Results

We compare our model with the following popular graph embedding learning methods and state-of-the-art drug response prediction methods.

- **Deepwalk** [38] first learns the embeddings for drugs and cell lines in the response-related network via

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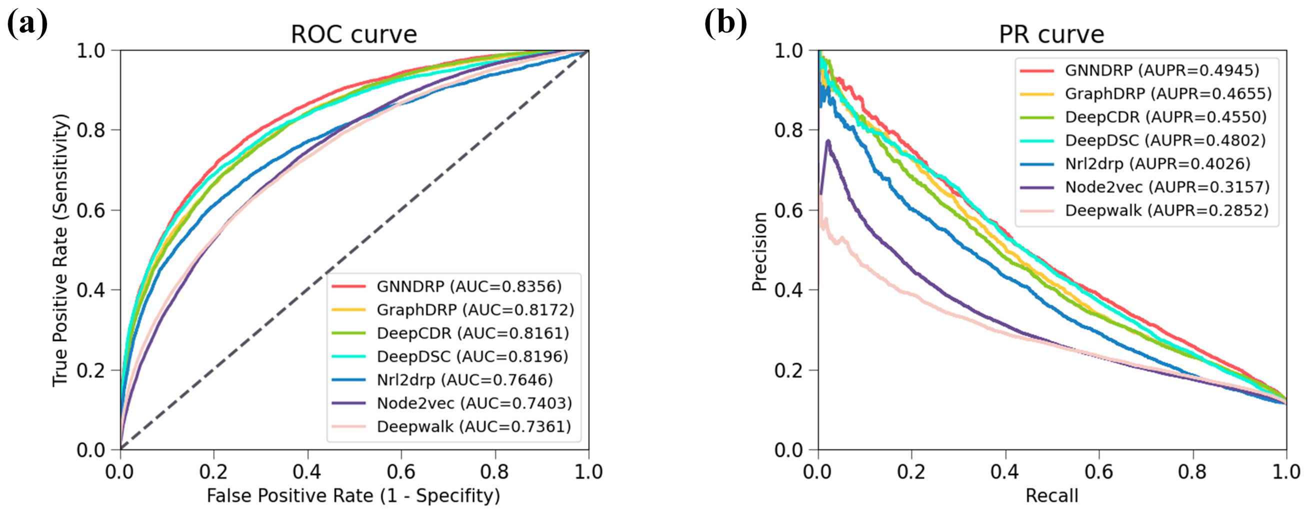

Fig. 2. Performance comparison of GNNDRP with baseline methods under rediscovering known drug-cell line response setting. (**a**) and (**b**) show the receiver-operating characteristic curve (ROC) curves and Precision-Recall (PR) curves of comparing methods, respectively.

random walk. Then, it predicts the relation for drug-cell line pairs via a linear layer.

- **Node2vec** [39] learns the embeddings for drugs and cell lines in the network. Similar to Deepwalk, it predicts the relation through a linear layer.
- **NRL2DRP** [9] constructs a large response-related network. Then, it uses LINE for the embedding learning of network nodes and applies SVM as a classifier to classify the binary relationship.
- **DeepDSC** [22] has two steps for predicting drug-cell line response. It firstly applies a stacked deep autoencoder to extract the latent representation of cell lines, and then combines the drug fingerprint to produce final response results.
- **DeepCDR** [24] extracts molecular graph embedding of drugs using a UGCN model and considers multi-omics profiles (genomic, transcriptomic and epigenomic) with different neural networks for predicting drug-cell line response.
- **GraphDRP** [25] applies various GNN models to process the SMILES sequences of drugs and uses CNNs to process cancer cell line's molecular profiles.

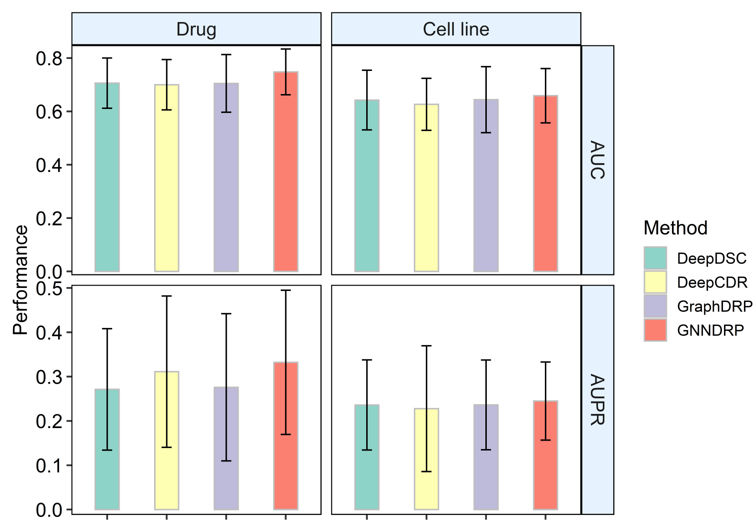

Fig. 3. Performance comparison under the blind test with unseen drug setting (left) and unseen cell line setting (right).

3.2.1 *Rediscovering known drug-cell line responses*

We use the receiver-operating characteristic (ROC) curve to visualize predictive performance and the AUC score to summarize the ROC curve. Since the resistant responses are much more than sensitive ones in our datasets (around 7.6:1) and the precision-recall (PR) curve is more informative than the ROC curve when evaluating binary classifiers on an imbalanced dataset [40], we introduce the PR curve and AUPR value to further assess the performance of our method. Fig.2(a) and (b) show the ROC curves and PR curves of all comparing methods, respectively. It is obvious that the ROC curve and PR curve of our method lies above the other six methods, demonstrating the effectiveness of GNNDRP on capturing the drug-cell line interaction information. Generally, deep learning-based methods (DeepDSC, DeepCDR, GraphDRP, and GNNDRP) significantly exceed other baselines. Among the four deep learning-based methods, GNNDRP outperforms best with a relatively large margin by achieving an AUC score of 0.8356 as compared to 0.8161 of DeepCDR, 0.8196 of DeepDSC, and 0.8172 of GraphDRP. This conclusion is also consistent considering the AUPR metric.

These results clearly verify the efficacy of GNNDRP. The substantial improvement of our model over the baselines could be credited to two reasons: our model integrates features extracted from the drug molecular graph, cell line genomic mutation state, and drug-cell line response network to predict drug response, and our model uses a self-supervised mechanism to enhance the performance of graph encoder.

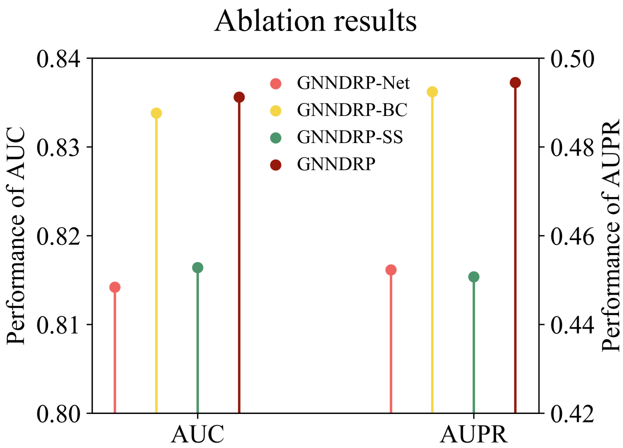

Fig. 4. Performance comparison of GNNDRP with different variants under rediscovering known drug-cell line response setting.

3.2.2 Blind test for both drugs and cell lines

In this section, we conduct the blind test for both drugs and cell lines. The prediction task becomes much more challenging since the drugs or cell lines in the test data are unseen during the training stage. The network-based methods (Deepwalk, Node2vec, and NRL2DRP) rely on known drug-cell line relationships to train the model, therefore these methods are inapplicable in the blind test setting. We compare GNNDRP with the other three baseline models that have performed well in the previous experiments and are available for the blind test setting. Fig.3 shows the predictive performance for the blind test with drugs and cell lines. Our proposed method GNNDRP reaches the best performance than other methods in the blind test for both drugs and cell lines with an average AUC of 0.7482 and AUPR of 0.3323 in drug blind test and an average AUC of 0.6589 and AUPR of 0.245 in cell line blind test.

For both the drug and cell line blind test, the predictive performances of all methods have a significant decline to that in rediscovering known drug-cell line response setting. This result suggests that predicting the drug-cell line response for unseen drugs or unseen cell lines is more difficult, but GNNDRP still achieves comparable performances when compared with other baseline methods. Considering drugs may share similar molecular structure information while genomic mutation states of cell lines are diversified, the blind test for drugs is slightly better compared with the blind test for cell lines.

3.3 Model ablation analysis

GNNDRP integrates both drug-cell line response network and biochemical characteristics of drugs and cell lines. Moreover, the self-supervised mechanism is also adopted to enhance the capability of the graph encoder and further boost the predictive performance. To evaluate the impact of these substructures of our model, we design several variants of GNNDRP and compare GNNDRP with these variants under rediscovering known drug-cell line response setting as follows:

- **GNNDRP-Net.** Without the network-based module, i.e. using the GIN module for drug SMILES data and CNNs module for cell line genomic data, respectively.
- **GNNDRP-BC.** Without the biochemical characteristic-based module, i.e. using the graph encoder to process the heterogeneous network and self-supervised learning strategy.
- **GNNDRP-SS.** Without the self-supervised mechanism, i.e. removing the self-supervised task while keeping the other module unchanged.

TABLE 1
Top 10 novel drug-cell line pairs predicted by GNNDRP

Drug	Cell line	Evidence
CX-5461	KASUMI-1	PMID: 28283481
Y-39983	KARPAS-231	NA
Trametinib	NK-92MI	NA
GSK1070916	SU-DHL-16	PMID: 19567821
Quizartinib	EoL-1-cell	PMID: 28895560
Alectinib	MV-4-11	NA
Trametinib	AGS	PMID: 32444897
GSK1070916	BL-41	NA
Nutlin-3a (-)	SIG-M5	NA
OSI-930	EoL-1-cell	NA

The ablation results are shown in Fig.4. We can see that the performance of GNNDRP-Net is significantly worse. This result indicates that the network-based module is effective in capturing interactions between drugs and cell lines and clearly increases the predictive performance. Notably, GNNDRP-BC obtains slightly lower performance than GNNDRP, indicating that biochemical characteristics of both drugs and cell lines can act as a supplement to the drug response information and also decisively to predict drug response. We choose the multi-task learning framework to achieve the self-supervised mechanism. When removing the self-supervised task, i.e. GNNDRP-SS only considers the drug response prediction task, the AUC score drops by 2.14% and the AUPR score degrades by 4.38%. This demonstrates that our self-supervised mechanism can significantly enhance the prediction ability.

3.4 Case study

In this section, our module GNNDRP is trained on all the known drug-cell line interactions and predicts the responses which are not included in our database. Because all known associations have been used to construct the prediction model, the predicted associations require verification by public literature or other available sources.

The top 10 drug-cell line interactions predicted by our models are listed in Table 1, and we can find evidence to confirm four out of them. For example, CX-5461 is capable of treating aggressive hematologic malignancies and KASUMI-1 is acute myeloid leukemia (ALL) cell line [41]. According to [42], GSK1070916 has antitumor effects in 10 human tumor xenograft models including breast, colon, lung, and two leukemia models, and they record the response of cell line SU-DHL-16 to GSK1070916 is 4nm (smaller than the threshold of GSK1070916).

Furthermore, we also take two clinically approved drugs Dasatinib and GSK690693 as examples. The top 10 ranked predicted sensitive cell lines of Dasatinib and GSK690693

TABLE 2
Top 10 scoring predicted sensitive cell lines of two drugs

Drug	Cell line	Evidence
Dasatinib	SK-N-SH	NA
	SU-DHL-5	NA
	RCH-ACV	PMID: 23153538
	BL-41	NA
	CAL-51	PMID: 17268817
	MHH-ES-1	NA
	SU-DHL-10	PMID: 29567799
	CRO-AP2	NA
	MC116	NA
	786-0	PMID: 26984511
GSK690693	RCH-ACV	PMID: 19064730
	HGC-27	NA
	CRO-AP2	NA
	KATOIII	PMID: 28860825
	GA-10	NA
	STS-0421	NA
	LB647-SCLC	NA
	MOLT-16	PMID: 19064730
	NCI-H929	NA
	HCC202	PMID: 26181325

are illustrated in Table 2, respectively. For Dasatinib, four cell lines, RCH-ACV [43], CAL-51 [44], SU-DHL-10 [45] and 786-0 [46] are identified to be sensitive by reporting experimental evidence. For example, SU-DHL-10 as the DLBCL cell line is sensitive to Dasatinib, since Dasatinib can block several SRC kinases concurrently and are therapeutically effective in lymphoma cells in vitro, irrespective of their molecular subtypes [45]. According to Roseweir et al.'s research [46], Dasatinib shows significant dose dependent influence on early nonmetastatic 786-0 ccRCC cell lines, increasing their apoptosis while significantly decreasing proliferation and migration. For the other drug GSK690693, four cell lines, RCH-ACV [47], KATOIII [48], MOLT-16 [47] and HCC202 [49], are supported to be sensitive. For instance, the relationship between gastric cancer (GC) cell line KATOIII and GSK690693 is reported in Lee et al.'s work [48]. In their study to investigate the sensitivity of the AKT inhibitor in ARID1A-deficient GC cells, they notice that GSK690693 increase apoptosis in ARID1A-deficient GC cells and record the IC50 value of KATOIII is 4.521μmol/L, which is smaller than the threshold of GSK690693.

To summarize, all the above outcomes demonstrate that GNNDRP can help to explore novel drug responses.

4 Conclusions

In this paper, we establish an end-to-end deep learning-based model, namely GNNDRP, which considers both biochemical characteristics and response-related network information. In addition, GNNDRP incorporates drug response prediction task and self-supervised task in a multi-task framework to further boost the predictive performance. We conduct a 5-fold cross-validation experiment on the benchmark dataset and compare it with the latest drug response prediction methods, proving that our proposed method for identifying drug responses is powerful and effective. Incorporating more drug-related and cell line-related information as well as other biological entities and relations into a knowledge graph to provide more meaningful information for drug response prediction will become part of our future research.

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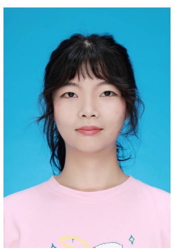

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